

Artificial Intelligence (AI) - Notes

Artificial Intelligence (AI) is a branch of computer science that focuses on creating machines capable of performing tasks that typically require human intelligence.

Key Concepts:

- Machine Learning (ML): Systems learn from data.
- Deep Learning: Neural networks with multiple layers.
- Natural Language Processing (NLP): Understanding human language.
- Computer Vision: Interpreting images and videos.

Applications:

- Chatbots (like ChatGPT)
- Self-driving cars
- Recommendation systems (Netflix, Amazon)
- Medical diagnosis

Advantages:

- Automation of repetitive tasks
- Faster decision making
- Improved accuracy

Challenges:

- Data dependency
- Bias in models
- High computational cost

Conclusion:

AI is transforming industries and will play a major role in the future of technology.